

$$B = \begin{pmatrix} \vec{b}_1 & \vec{b}_2 \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}$$

skorpee jian vektor base

$$B^T = \begin{pmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{pmatrix} = \begin{pmatrix} b_1^T \\ b_2^T \end{pmatrix}$$

$$\langle x, y \rangle = x^T A y$$

$$\langle b_i, b_j \rangle = b_i^T A b_j$$

$$(B^T A B)_{ij} = b_i^T A b_j$$

$$\begin{aligned} \langle b_1, b_1 \rangle &= 1 \\ \langle b_2, b_2 \rangle &= 1 \\ \langle b_1, b_2 \rangle &= \langle b_2, b_1 \rangle = 0 \end{aligned}$$

pro ON bazi $\langle b_i, b_j \rangle = \delta_{ij}$

využit se gramovy matice

$$I_2 = B^T A B = \begin{pmatrix} \langle b_1, b_1 \rangle & \langle b_1, b_2 \rangle \\ \langle b_2, b_1 \rangle & \langle b_2, b_2 \rangle \end{pmatrix} = I_2$$

vyjádrení A:

$$A = (B^T)^{-1} B^{-1} = (B B^T)^{-1}$$

$$B \cdot B^T = \begin{pmatrix} 1 & 3 \\ 2 & 7 \end{pmatrix} \cdot \begin{pmatrix} 12 & 37 \\ 23 & 53 \end{pmatrix} = \begin{pmatrix} 10 & 23 \\ 23 & 53 \end{pmatrix}$$

$$\det(B \cdot B^T) = 1$$

A^{-1} pomocí adjungované matice

$$(B \cdot B^T)^{-1} = \begin{pmatrix} 53 & -23 \\ -23 & 10 \end{pmatrix} = A$$

$$\langle \vec{x}, \vec{y} \rangle = \vec{x}^T A \vec{y} = 53x_1y_1 - 23x_1y_2 - 23y_1x_2 + 10x_2y_2$$